

Machine Learning Algorithms

- Linear Regression
- Logistic Regression
- Naive Bayes
- SVM
- KNN
- Decision Tree
- Random Forest
- Gradient Boosting
- XGBoost

NLP Core

- Tokenization
- N-gram
- Bag of Words
- TF-IDF
- Word2Vec
- GloVe
- FastText
- Text Preprocessing

Deep Learning

- Perceptron
- MLP
- Backpropagation
- RNN

- LSTM
- GRU
- Seq2Seq
- Attention Mechanism

Transformer / Modern NLP

- Self-Attention
- Multi-Head Attention
- Positional Encoding
- Encoder-Decoder Architecture
- Masking
- BERT (by Google)
- GPT (by OpenAI)
- LLaMA (by Meta)
- Fine-tuning
- LoRA
- RAG

Frameworks & Libraries

 **Python**

 **NumPy**

 **Pandas**

 **Matplotlib**

 **Scikit-learn**

 **PyTorch**

 **HuggingFace Transformers**

🔗 Git

🔗 SQL

◆ **DSA (Big Tech Requirement)**

- Array
- String
- Hashmap
- Stack
- Queue
- Linked List
- Tree
- Heap
- Graph
- Recursion
- Dynamic Programming
- Time & Space Complexity

◆ **Extra (Internship + Job Edge)**

- Model Evaluation Metrics
- Cross Validation
- Hyperparameter Tuning
- ML System Design
- Basic MLOps
- API Deployment
- Linux Basics
